

Preface

“UX” MEANS USER EXPERIENCE

Welcome to *The UX Book*, the second edition. We thought we’d better begin by making sure everyone knows that “UX” is short for user experience. In simple terms, the user experience is the phenomenon felt by users before, during, and after usage—usually a combination of usability, usefulness, emotional impact, and meaningfulness.

GOALS FOR THIS BOOK

Imbue an understanding of what a good user experience is and how to achieve it. Our main goal for this book is simple: to help readers learn to recognize, understand, and design for a quality user experience (UX). Sometimes, a quality user experience is like an electric light: when it works, nobody notices. Sometimes, a user experience is really good and is noticed and even appreciated, and it leaves pleasant memories. Or sometimes, the effects of a bad user experience persist in the user’s mind. So, early in the book, we talk about what constitutes a positive user experience.

Emphasize design. What is less obvious than understanding a quality user experience is how to design for it. Perhaps the most significant change in this edition is that we stress design—a kind of design that highlights the designer’s creative skills and insights and embodies a synthesis of technology with aesthetics and meaningfulness to the user. In [Part 3](#), we show you a variety of design methods to help you get just the right approach for your project.

Give a how-to approach. We have designed much of the book as a how-to-do-it handbook and field guide, as a textbook for students aspiring to be UX professionals and for professionals aspiring to be better. The approach is practical, not formal or theoretical. Some references are made to the related science, but they are usually to provide context for the practice and are not necessarily elaborated.

Other goals for the reader. Beyond our main goal of helping readers learn about UX and UX design, other goals for the reader experience include ensuring that:

- You get excited about UX design.
- The material is easy to learn.
- The material is easy to apply.
- The material is useful to students and professionals.
- For you, the reader, the experience is at least a little bit fun.

Provide broad coverage. Our goals for scope of coverage include:

- Depth of understanding—detailed information about different aspects of the UX process (like having an expert accompanying the reader)
- Breadth of understanding—as comprehensive as space permits
- Broad range of application—the process and the design infrastructure and vocabulary, including guidelines, are not just for GUIs and the web, but for all kinds of interaction styles and devices, including ATMs, refrigerators, road signs, ubiquitous computing, embedded computing, and everyday things.

USABILITY IS STILL IMPORTANT

The study of usability, a key component of a quality user experience, is still an essential part of the broad and multidisciplinary field of human-computer interaction. It is about getting our users past the technology and focusing on getting things done. In other words, it is about designing the technology as an extension of human capabilities to accomplish something, and to be as transparent as possible in the process.

BUT USER EXPERIENCE IS MORE THAN USABILITY

As our discipline has evolved and matured, more and more technology companies are embracing the principles of usability engineering, investing in sophisticated usability labs and personnel to “do usability.” As these efforts are becoming effective at ensuring a certain level of usability in the products (thereby, leveling the field on that front), new factors have emerged to distinguish competing product designs.

We will see that, in addition to traditional usability attributes, user experience entails social and cultural, value-sensitive design, and emotional impact—how the interaction experience includes “joy of use,” fun, aesthetics, and meaningfulness in the lives of users.

The focus is still on designing for the human rather than on technology, so, “user-centered design” is still a good description. But it is now extended to knowing users in newer and broader dimensions.

A PRACTICAL APPROACH

This book takes a practical, applied, hands-on approach, based on the application of established and emerging best practices, principles, and proven methods to ensure a quality user experience. Our approach is about practice, drawing on the creative concepts of design exploration and visioning to make designs that appeal to the emotions of users, while moving toward processes that are lightweight, rapid, and agile—to make things as good as resources permit, and not waste time and other resources in the process.

PRACTICAL UX METHODS

In the first edition, we described mostly rigorous methods and techniques for each UX lifecycle activity, mentioning more rapid methods here and there. UX designers who need rigorous methods to develop large-scale systems in complex domains will still find what they need in this book. But this edition reflects the fact that agile methods have taken on a much bigger role in UX practice. We have adopted a practical stance with respect to rigor and formality, and our processes, methods, and techniques are a practical compromise between rigor and speed that will suit a very large percentage of all projects.

FROM AN ENGINEERING ORIENTATION TO A DESIGN ORIENTATION

For a long time, the focus in HCI practice was on engineering, taking its cues from usability engineering and human factors engineering. Our first edition mostly reflected this approach. In this new edition, we are moving away somewhat from an engineering focus to more of a design focus. In an engineering-centric view, we started with constraints and tried to design something to fit those constraints. Now, in a design-centric view, we envision an ideal experience and then try to push the technology limits to make it happen and realize the vision.

AUDIENCES

Anyone involved in, or wishing to learn more about, creating products that engender a quality user experience will benefit from this book. One important audience is *students and the instructors* who teach them. Another important intended audience includes UX practitioners: *UX professionals* or others who take on the role of a UX professional in project environments. The perspective of professionals is very similar to that of students; both have the goal of learning, only in slightly different settings and perhaps with different motivations and expectations.

Our readership comprises all kinds of UX professionals, including UX designers, content strategists, information architects, graphic designers, web designers, usability engineers, mobile device app designers, usability analysts, human factors engineers, cognitive psychologists, cosmic psychics, trainers, technical writers, documentation specialists, marketing personnel, and project managers. Readers in any of these areas will find the hands-on approach of this book valuable and can focus mainly on the how-to-do-it parts.

Software people who work with UX professionals—including software engineers, programmers, systems analysts, software quality-assurance specialists—can also benefit from this book. Software engineers who find themselves called upon to do some UX design will find this book easy to read and apply because UX design lifecycle concepts are similar to those in software engineering.

WHAT'S CHANGED SINCE THE FIRST EDITION?

Sometimes, when you set out to write a second edition, you end up essentially writing a new book. That is the case with this edition. So much has changed since the first edition, including our own understanding of, and experience with, the process. Porter put it this way, long ago: “The present work on the Health, Pleasures, Advantages, and Practice of Cycling is based largely on the author’s previous writings on the same subjects, and principally on a volume bearing the same name which he published in 1890. But the changes that have taken place since the other work was written have been so great that no mere revision, but a complete rewriting proved necessary, in order to eliminate portions that were out of date, and to add very much that was new and important.” (Porter, 1895).

NEW CONTENT AND EMPHASIS

Here are some of the new topics or approaches to material you will find in this edition:

- Increased focus on design. Many of the process-oriented chapters are infused with a new emphasis on design, design thinking, and generative design. We even changed the subtitle of the book slightly to reflect this emphasis (to *Agile UX Design for a Quality User Experience*).
- New approach to processes, methods, and techniques. The early chapters establish process-related terminology and concepts as context for discussion in later chapters.
- The entire book is oriented toward an agile UX lifecycle process as a better match to the now *de facto* standard agile approach to software engineering. We have also introduced a model (the funnel model of agile UX) to explain the role of UX in various different development environments.
- Commercial product perspective and enterprise system perspective. These two quite different kinds of UX design settings are now explicitly recognized and treated separately.

TIGHTENED UP THE VERBOSE TEXT

We got feedback on the first edition from people who want us to tighten up the writing. To make the second edition easier to read, we tried to make it concise by eliminating repetition and verbosity. We hope you will agree that we did our best at addressing this.

A MORE RELAXED APPROACH TO GRAMMAR AND WRITING STYLE

To enhance readability, we have adopted some changes in the mechanics of writing. We moved toward a more informal, and hopefully more readable, style by relaxing some grammatical rules. Good grammar is important. However, some of the old grammatical constraints make the writing and reading awkward. For example, many people now feel that split infinitives are often easier on the ear (to more casually communicate) than the more formal alternative.

Most of those rules were never real, anyway. And, in case you are a stickler about these “rules,” they may not be what you think. It’s not that the language has

evolved through modern usage to make the breaking of these rules the new norm. It's more that these so-called rules never were real (O'Conner & Kellerman, 2013, p. 24). And, beyond that, (for the *real* sticklers) there is nothing to split in an infinitive, because the word "to" is technically not part of the infinitive. Famous authors such as Shakespeare, Donne, and Wordsworth have been "splitting infinitives" since the 1200s. So, we hope we won't be faulted for easing up on some grammatical rules that some have thought to be ironclad, but which are really optional. See also Pinker's book, *The Sense of Style* (Pinker, 2014).

How we are relaxing the grammar. We may occasionally allow a sentence to start with a conjunction. And that's perfectly good when it's needed. We sometimes find a preposition a good thing to end a sentence with. We're also now in the habit of using contractions. By some standards, they are verboten in published writing, but, you have to agree, informal speech flows nicely with contractions, and we see no reason you can't experience that same pleasant cadence in your reading.

Another example of something you're not supposed to do is have a one-sentence paragraph, but sometimes it's a nice way to highlight a point.

In addition, we use personal pronouns, a practice not usually admired in formal writing but, well, we're trying to be less formal. So, instead of saying, "the reader will notice...", we'll use the less formal and friendlier, "you will notice..." Similarly, we'll use "we" or "us" to refer to ourselves, the authors. Sometimes, we'll say "I" or "me," and you might have to guess which of us that means.

Another area where we will be relaxed is anthropomorphic wording, which might cause some grammarians to roll over in their ivory towers, but sometimes, it's just easier, and readers will always know what we mean. For example, we might say, "Figure X shows such and such," when, of course, we mean, "in Figure X, we show such and such." I'll bet that, if we hadn't mentioned that one here, most readers wouldn't have even noticed.

Gender-specific pronouns. We have observed that literary acrobatics to avoid the use of gender-specific pronouns can lead to considerable stiffness in writing. We are saddled with a language that has built-in gender traps that are difficult and awkward to avoid. Sometimes, you can just use the plural—for example, instead of "the user must type Ctrl-Shift-Alt-Delete often, and he will find that difficult," you can say, "users must type Ctrl-Shift-Alt-Delete often, and they will find that difficult." At least that is better than the dreadful mix of number in "the user must type Ctrl-Shift-Alt-Delete very often, and they will find that difficult." Other approaches to de-gendering can result in graceless, lashed-together combinations such as he/she or s/he or the ungainly and mysterious "one," as in "When one takes on the role of UX professional, then one must be very brave."

We solve this gender trap by hitting it head on. We use “he” or “she” when useful and will try to stay out of political trouble by using each gender equally often. Can you tell if we are counting?

Relaxing about references. Most of the references cited will be in a standard academic-style. But sometimes, a less formal source is sufficient and practical. In particular, occasionally we’ll use a reference to Wikipedia for a “definition” of a commonly-used term. Yes, sometimes Wikipedia and other similar sources can be flawed, but we will do our best to avoid anything questionable.

Also this. And we shall struggle to never use the word “plethora.”

WHAT WE DON’T COVER

To begin with, this book is not a survey of the HCI field, nor is it a survey about user experience. It is also not about human-computer interaction research. Although this book is large and comprehensive, we could not hope to include everything about HCI or UX. We apologize if your favorite topic is excluded, but we had to draw the line somewhere. Further, many of these additional topics are so broad in themselves that they could (and most do) fill a book of their own.

Among the specific topics not included are:

- Accessibility, special needs, and the American Disabilities Act (ADA)
- Internationalization and cultural differences
- Standards
- Ergonomic health issues, such as repetitive stress injury
- Specific HCI application areas, such as societal challenges, healthcare systems, help systems, training, and designing for elders or other special user populations
- Special areas of interaction such as virtual environments or 3D interaction
- Computer-Supported Collaborative Work (CSCW)
- Social media
- Personal information management (PIM)
- Sustainability (we had originally planned to include this, but space ran out)
- Summative UX evaluation studies

ABOUT THE EXERCISES

A hypothetical system called the Ticket Kiosk System is used as a UX design example to illustrate the material in throughout the process chapters. In this running example, we describe activities that you can imitate to build your own designs. The exercises are an integral part of the learning from this book. In its

use of hands-on exercises based on the Ticket Kiosk System, the book is somewhat like a workbook. After each main topic, you get to apply the new material immediately, learning the practical techniques by active engagement in their application. The book is organized and written to support active learning (learning by doing), and should be used that way.

The exercises require medium-level engagement, somewhere in between the in-text examples and full project assignments.

Take them in order. Each process chapter builds on the previous ones and adds a new piece to the overall puzzle. Each exercise builds on what you learned and accomplished in the previous stages—just as in a real-world project.

Do the exercises as a team if you can. Good UX design is almost always a collaborative effort. Working through the exercises with at least one other interested person will enhance your understanding and learning of the materials greatly. In fact, many of the exercises are written for small teams (for example, three to five people) because they involve multiple roles.

The teamwork will help you understand the kinds of communication, interaction, and negotiation that take place in creating and refining a UX design. If you can season the experience by including a software developer with responsibility for software architecture and implementation (at least for a working prototype), many important communication needs will become apparent.

Students should do the exercises as team in the classroom. If you are a student in a course, the best way to do the exercises is as team-based in-class exercises. The exercises are adapted easily for classroom use as an ongoing, semester-long interactive set of in-class activities to understand needs, design solutions, prototype candidate designs, and evaluate the UX. The instructor can observe and comment on your progress, and you can share your “lessons learned” with other teams.

UX professionals should get buy-in to do the exercises at work. If you are a UX professional, or you aspire to learn on-the-job to be a UX professional, trying to learn this material in the context of your regular work, the best way of all is an intensive short course with team exercises and projects. We used to teach just such a short course.

Alternatively, if you have a small UX team in your work group (perhaps a team that expects to work together on a real project), and your work environment allows, set aside some time (say, two hours every Friday afternoon) for the team exercises. To justify the extra overhead to pull this off, you might have to convince your project manager of the value added.

Individuals can still do the exercises. Do not let the lack of a team stop you from doing the exercises. Try to find at least one other person with whom you can

work, or, if necessary, get what you can from the exercises on your own. Although it would be easy to let yourself skip the exercises, we urge you to do as much on each of them as your time permits.

TEAM PROJECTS

Students. In addition to the small-scale series of in-class activities in conjunction with the book exercises, we offer fully detailed and more involved team projects. We consider this semester-long sequence of team projects an essential part of learning by doing in a course based on this book. These team projects have always been the most demanding, but also the most rewarding, learning activity in the course.

For this semester-long team project, we use real clients from the community—a local company, store, or organization that needs some kind of interactive software application designed. The client stands to get some free consulting and even (sometimes) a system prototype in exchange for serving as the project client. A sample set of team project assignments is available in the Instructor's Guide, available to instructors from the publisher.

UX professionals. As a way of getting started in applying this material to your real work environment, you and your coworkers can select a low-risk, but real, project. Your team may already be familiar and even experienced with some of the activities we describe and may even already be doing some of them in your development environment. By making them part of a more complete and informed development lifecycle, you can integrate what you know with new concepts presented in the book.

ABOUT THE AUTHORS

Rex Hartson is a pioneer researcher, teacher, practitioner, and consultant in HCI and UX. He is the founding faculty member of HCI (in 1979) in the Department of Computer Science at Virginia Tech. With Deborah Hix, he was co-author of one of the first books to emphasize the usability engineering process, *Developing User Interfaces: Ensuring Usability Through Product & Process*. Hartson has been principal investigator (PI) or co-PI at Virginia Tech on a large number of research grants and has published many journal articles, conference papers, and book chapters. He has presented many tutorials, invited lectures, workshops, seminars, and international talks. He was editor or co-editor for *Advances in*

Human-Computer Interaction, Volumes 1–4, Ablex Publishing Co., Norwood, New Jersey. His HCI practice is grounded in over 30 years of consulting and UX Design training for dozens of clients in business, industry, government, and the military.

Pardha S. Pyla is an award-winning designer and strategist with deep expertise in envisioning and delivering industry-leading products. He is a seasoned leader with a record of establishing exemplary user experience and product development practices. He is a pioneering researcher in the area of coordinating software engineering and UX lifecycle processes and the author of several peer-reviewed research publications in human-computer interaction and software engineering. He has received numerous awards in recognition of his work in design, research, teaching, leadership, and service.